DAY 01

RED TEAMING FUNDAMENTALS

> **Think Like an Adversary.**
> **Operate Like a Professional.**

From reconnaissance to objectives — understanding how real red teams approach security assessments, one operational concept at a time.

INTRODUCTION

Welcome to the 10-Day Red Team Series

This series is designed to introduce you to the mindset, methodology and practical workflow used during authorized red-team operations — the same way a new operator would be brought up to speed inside a real security team.

The goal of this series is **not** simply to teach you tools. Tools change constantly. What stays constant is how a skilled operator thinks, plans and moves through an environment. Over the next ten days, you'll build that foundation piece by piece.

By working through this series, you will learn to:

- ☑ Think from an adversary's perspective
- ☑ Identify realistic attack paths
- ☑ Understand how individual weaknesses connect into a chain
- ☑ Move from initial access toward a defined objective
- ☑ Document findings the way a professional operator would
- ☑ Understand how a defender would see the same activity
- ☑ Practice safely inside authorized environments only

> **ON OBJECTIVES**
>
> **The goal of a red team is not to find as many vulnerabilities as possible. The goal is to demonstrate realistic attack paths and determine what an attacker could actually achieve.**

DAY 01 OBJECTIVE

By the end of Day 1, you should understand:

- What red teaming is
- How it differs from penetration testing
- The red-team mindset
- The typical attack lifecycle
- Rules of engagement
- How red team operations are structured

CORE CONCEPT

What Is Red Teaming?

A red team simulates realistic adversary behavior against an organization within an authorized scope.

Rather than checking systems off a list, a red team operation is built around a small set of hard questions — the same questions a real attacker, and a real defender, both care about:

- ☑ Can an attacker gain initial access?
- ☑ Can they escalate privileges once inside?
- ☑ Can they move through the environment undetected?
- ☑ Can they reach and access sensitive systems?
- ☑ Can security controls detect and respond to the activity?
- ☑ Can the attacker reach the defined objective?

Red teaming focuses on **attack paths and objectives** — not simply on producing a long list of vulnerabilities. A finding only matters as much as the path it opens up.

The Attack Path Model

A simplified view of how isolated weaknesses turn into operational impact.



WHY THIS MODEL MATTERS

A single weakness rarely matters in isolation. It matters because of what it connects to. Professional operators evaluate every finding by asking: *what does this let me reach next?*

DISTINCTIONS

Red Teaming vs Penetration Testing

These terms are often used interchangeably — but they describe different engagement styles, with different goals and different definitions of success.

	VULNERABILITY ASSESSMENT	PENETRATION TEST	RED TEAM
Primary goal	Identify known weaknesses	Exploit weaknesses to prove risk	Achieve a real-world objective
Scope	Broad, systems-based	Defined system or application	Whole organization / attack surface
Attack realism	Low	Moderate	High — mimics real adversaries
Stealth	Not a factor	Rarely a factor	Central to the engagement
Objectives	Coverage of known issues	Validated vulnerabilities	Defined business-impact goal
Duration	Short (days)	Short–medium (1–3 weeks)	Extended (weeks–months)
Human interaction	Minimal	Some	Actively evades analysts & controls
Lateral movement	Not tested	Sometimes tested	Core component
Detection testing	No	Limited	Yes — tests people & process too
Reporting	Vulnerability list	Findings + exploitation proof	Attack narrative + path + impact

Important

Red teaming and penetration testing overlap, but they are not identical.

A penetration test may focus heavily on identifying and validating vulnerabilities within a defined scope. A red team operation focuses more heavily on demonstrating whether an adversary can achieve a meaningful objective through a realistic — and often stealthy — attack path.

THE MOST IMPORTANT PAGE TODAY

Think Like an Adversary

There is a real difference between a tool-first mindset and an objective-first mindset — and it's the single biggest tell between a beginner and a professional operator.

Tool-First Thinking

"What can I exploit with this tool?"

VS

Objective-First Thinking

"What does the target protect, and what path could realistically lead me there?"

Professional operators constantly ask themselves

- ☑ What am I trying to achieve?
- ☑ What information do I currently have?
- ☑ What access do I have?
- ☑ What assumptions am I making?
- ☑ What is my next best move?
- ☑ What would the defender see?
- ☑ What evidence am I leaving behind?
- ☑ What alternative attack paths exist?

> **REMEMBER**
>
> **Tools are replaceable. Methodology is not.**

METHODOLOGY

From Reconnaissance to Objective

The stages a red team operation typically moves through, in rough order.

1. **Reconnaissance**
   Gathering information about the target before any interaction that could be noticed.

2. **Initial Access**
   Establishing the first foothold into the target environment.

3. **Execution**
   Running code or commands to carry out an action on the compromised system.

4. **Persistence**
   Maintaining access across reboots, credential changes and time.

5. **Privilege Escalation**
   Gaining higher-level permissions than the initial foothold allowed.

6. **Credential Access**
   Obtaining account names, passwords, tokens or keys.

7. **Discovery**
   Learning the internal layout — systems, trust relationships and defenses.

8. **Lateral Movement**
   Moving from one system to another toward the objective.

9. **Objective**
   Reaching and demonstrating impact on the defined target asset.

10. **Cleanup & Reporting**
    Removing operational artifacts and documenting the full engagement.

> **NOTE**
>
> **A red team operation is an adaptive process, not a checklist. Real operations are rarely linear — operators move backward and forward between stages as new information becomes available.**

DIRECTION

What Are You Actually Trying to Achieve?

Every legitimate red-team engagement should have a clearly defined objective — a concrete outcome the operation is built around, agreed with the client in advance.

Typical engagement objectives

- ☑ Demonstrate access to a sensitive internal system
- ☑ Determine whether an attacker could reach a critical server
- ☑ Demonstrate compromise of a specific business asset
- ☑ Test detection and response capabilities
- ☑ Determine whether an external attacker could reach an internal objective
- ☑ Assess the effectiveness of security controls across an attack path

▲ FINDING A VULNERABILITY ≠ ACHIEVING THE OBJECTIVE

A vulnerable web server may be interesting. But if the engagement objective is to determine whether an attacker can reach a specific internal asset, the operator must understand how that vulnerability actually contributes to the broader attack path — not just that it exists.

EXAMPLE

An outdated CMS plugin on a public marketing site is a real finding. On its own, it says little about business risk. The operator's job is to ask what that access leads to next — internal network reachability, stored credentials, or nothing at all — and report accordingly.

GOVERNANCE

Rules of Engagement

Red teaming must be authorized and carefully controlled. Before any technical activity begins, a professional engagement defines exactly what is — and isn't — permitted.

- **Scope** — which domains, IP addresses, applications, users and systems are included
- **Authorization** — who has approved the operation, in writing
- **Testing window** — when testing is allowed to occur
- **Restrictions** — what actions are explicitly prohibited
- **Safety** — which systems or services must never be disrupted
- **Communication** — who to contact if something goes wrong
- **Stop conditions** — when the operation must immediately stop
- **Evidence** — what can be collected and how it must be protected

> ▲ **AUTHORIZED ENVIRONMENTS ONLY**
>
> Red-team techniques should only ever be performed against systems you own, or have explicit written authorization to assess. Every exercise in this series assumes a fictional company, a lab environment, or an explicitly authorized target.

APPLIED SCENARIO — FICTIONAL ORGANIZATION

Mission Brief: Operation GreenFox

You have been authorized to perform a controlled red-team assessment against a fictional target: **GreenFox Industries.**

OBJECTIVE

Determine whether an external attacker could gain access to a designated internal business system.

Available information

- Organization name
- Public domain
- Approved external IP range
- Testing window
- Defined objective
- Emergency contact

Your first questions

1. What information would you collect first?
2. What assets would you identify?
3. What would you avoid touching?
4. How would you document your findings?
5. What would define success?
6. What would make you stop the operation?

THINK BEFORE YOU ATTACK

The purpose of this exercise is to develop methodology before introducing tools. Answers aren't provided here — sit with the questions.

PRACTICE

A Simple Operator Workflow

1. Understand the Objective
2. Map the Attack Surface
3. Identify Potential Paths
4. Validate Access
5. Expand Carefully
6. Escalate Where Authorized
7. Move Toward the Objective
8. Collect Evidence
9. Document Everything
10. Report

Why documentation matters

An action that isn't documented didn't happen — not for reporting purposes, and not for the client trying to reproduce or fix it. Every meaningful action during an operation should be logged with:

`timestamp` `target` `action performed` `result` `evidence` `security impact` `next step`

EXERCISE

Day 01 Challenge

SCENARIO

You are given an authorized assessment of a fictional company's external environment. You know the company name, public domain and approved testing scope — but nothing else.

Challenge

Before using any exploitation tools, list the first **10 things** you would want to understand about the target.

YOUR NOTES

Then ask yourself

> **What would your first action be, and why?**

There is no single correct answer. Your reasoning matters more than the tool you choose.

SUMMARY

Day 01 — Key Takeaways

- [x] Red teaming is objective-driven
- [x] Red teams simulate realistic adversary behavior
- [x] Tools are only part of the process
- [x] Reconnaissance comes before exploitation
- [x] Attack paths matter more than isolated vulnerabilities
- [x] Rules of engagement protect both the client and the operator
- [x] Documentation is part of the operation, not an afterthought
- [x] Red teaming must always be authorized
- [x] A good operator constantly thinks about the defender's perspective

> CLOSING THOUGHT
>
> **Don't ask only “What can I hack?” Ask “What could an attacker achieve?”**

KEEP GOING

Don't Stop at Theory

Reading about red teaming is useful. Practicing it is where the skill actually develops.

Codelivly lets you practice cybersecurity through hands-on labs, learning paths, CTFs and practical challenges — built to reinforce exactly the kind of methodology covered in this series.

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DAY 01 COMPLETE ✓

NEXT UP

Tomorrow: Day 02 — Reconnaissance & OSINT

> **Before an attacker can exploit an organization, they need to understand it.**

Preview

- Passive reconnaissance
- Active reconnaissance
- DNS fundamentals
- Domains and subdomains
- Public information gathering
- Technology discovery
- Attack surface mapping
- OSINT methodology

Come back tomorrow. The operation begins with information.

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